Graduation Project Route

Sales & Marketing Analysis Dashboard

Submitted to :Eng\ibrahim Al-Nwasany

Submitted by:

1-mariam Mostafa 2-Basmala Taha

Project Documentation: Sales & Marketing Analysis Dashboard

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1. Project Overview

This project represents a **comprehensive Data Analysis and Visualization solution** developed using **Power BI**, based on a dummy dataset simulating real-world **sales, marketing, store, and product performance data**.
The primary objective of this project was to build **insightful, visually appealing dashboards** that support **strategic decision-making** for different organizational levels — from **C-Level executives** to the **sales operation teams**.

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2. Project Objective

The main goal of the project is to **transform raw data into clear, actionable insights** through interactive dashboards.
The dashboards are designed to:

- Present data in a **clean, structured, and creative way**.
- Support **strategic and operational decisions**.
- Enable management to **track performance against targets**.
- Highlight key business indicators such as **Sales, Orders, Profit, Quantity, Returns, Promotions, and Marketing Activities**.

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3. Dataset Overview

The dataset consists of **9 interconnected tables**, covering all aspects of sales and marketing operations:

Table Name	Description
Data	Contains transaction-level details of sales and orders.
Promotion	Includes data related to promotions and discounts.
Marketing	Tracks impressions, clicks, and campaign performance.

Table Name	Description
Store	Provides store-level information (type, size, employees).
Channel	Represents the different sales channels.
Geography	Defines regions, countries, and continents.
Product	Contains product-level details.
ProductCategory	Groups products into categories.
ProductSubCategory	Further divides categories into subcategories.

Each table contains multiple columns and was used to create relationships during the **data modeling** phase.

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4. Data Understanding & Exploration

The provided dataset was **dummy data** representing a global retail company.
During data exploration, relationships between tables were analyzed to understand data flow and business meaning.

Key points in this phase:

- Identified main dimensions: *Product, Geography, Channel, Store*.
- Identified facts: *Sales, Orders, Promotions, Marketing Performance.*
- Validated integrity of relationships and adjusted incorrect or missing ones.
- Defined key questions and hypotheses to guide dashboard creation, such as:
    - Which countries and stores achieve the highest profit?
    - How do returns affect total revenue?
    - Which promotions and marketing campaigns drive the most engagement?
    - How does performance vary across years and months?

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5. Data Modeling

A **Star Schema** model was built to connect all related tables efficiently.

- The **Fact table** (Sales Data) connects to **Dimension tables** (Product, Store, Geography, Promotion, Marketing, etc.).
- Relationships were established as *many-to-one*, ensuring correct data aggregation.
- Data model optimization included:
    - Adding **Date table** for time intelligence.
    - Creating **keys** between Product and its Category/Subcategory.
    - Linking **Store** with Geography and Channel for location-based insights.
    - Implementing **RLS (Row-Level Security)** for user access control:
        - *User A:* Access to all TV and Video categories.
        - *User B:* Access only to "Car Video" subcategory within "TV and Video".

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6. Dashboard Design

The project includes **two main dashboard types**:

1. C-Level Executive Dashboards

- **Sales Dashboard:**
  Provides an overview of key KPIs such as Total Sales, Profit, Quantity, Orders, and Returns.
  Helps executives assess overall company performance vs. targets and previous years.
- **Returns Dashboard:**
  Focuses on return trends, rates, and their impact on profit margins.

2. Sales Team Dashboards

Includes 6 pages:

- **Sales Page:** Monthly sales trends, YTD vs LY comparisons, and performance by region.
- **Return Page:** Return quantities, reasons, and impact on total sales.
- **Geography Page:** Sales and profit per country and continent, with drill-through functionality.

- **Time Page:** Yearly and monthly slicers, order delivery time analysis, and date-based comparisons.
- **Promotion Page:** Active vs inactive promotions, average discount %, and KPI tracking.
- **Marketing Page:** Clicks, impressions, CPC, and campaign performance over time.

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7. DAX Measures Created

More than **70 DAX measures** were developed to calculate KPIs, comparisons, and dynamic behavior.
Key DAX measures include:

📈 **Performance KPIs**

- Total Sales, Target Sales, and YTD Sales
- Total Orders, Target Orders, Orders YTD
- Total Quantity, Quantity Target, YTD vs LY YTD %
- Total Profit, Profit Target, Profit Margin %
- Overachieving vs Underachieving KPIs

🔁 **Return & Quality**

- Total Returns, Return Rate %, Return Impact
- Avg. Return Value per Product and Transaction

📊 **Operational Metrics**

- Sales per Employee
- Avg. Discount %
- Avg. Promotion Days
- Active Stores %
- CPC (Cost per Click)
- Average Clicks per Day

🌍 **Geography & Marketing**

- Total Countries, Continents, and Stores
- Top City by Profit, Top Country by Sales
- Active Promotions
- Clicks YTD, LY YTD, and Change %
- Time Range Change (15, 30, 45 days)

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8. Insights Summary

Sales & Orders

- Total Sales: **$8B**, Target: **$9.5B**, YTD: **$2.55B**
- Orders: **2M**, Target: **3M**, YTD: **605K**

Quantity & Returns

- Quantity: **37M**, Target: **35M**, YTD: **15M**
- Return Rate: **42% (YoY)**

Stores & Employees

- Total Stores: **306**, Employees: **47**
- Top Store Type: Store | Top Close Reason: Relocation

Geography

- 35 Countries | Top Country: United States | Top Continent: North America

Products & Promotions

- Products: **2K**, Brands: **10**
- Top Product: *Proseware Projector 1080p*
- Active Promotions: **5 / 28**, Avg. Discount: **13%**
- Top Clicks (2023): **9M**

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9. Recommendations

- ☑ **Boost promotions and campaigns** to increase sales and order conversions.

- ⚠️ **Investigate return causes (42%)** — possibly product quality or shipment issues.
- 👩💼 **Increase employee training** to improve sales per employee ratio.
- 🏷️ **Expand high-performing categories** like *Projectors & Screens*.
- 🌍 **Focus marketing on top-performing regions** — North America and the U.S.

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10. Dashboard Design Highlights

- Used consistent **color coding**, **icons**, and **shapes** for clarity.
- Implemented **tooltips, drill-through, bookmarks**, and **dynamic formatting**.
- Ensured every visual aligns with storytelling — from KPIs to trends.
- Enhanced readability through **professional alignment and spacing**.
- Maintained balance between analytical depth and visual simplicity.

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11. Conclusion

This project successfully demonstrates advanced analytical and visualization skills using Power BI.
The dashboards deliver meaningful insights that can directly **support strategic decisions** for different business levels.
By integrating clean data modeling, dynamic DAX calculations, and creative design, this project achieves both **technical excellence** and **business relevance**.